

Esercizio.

For every set X , $Fn(X, 2)$ has the CCC.

Dimostrazione.

Let $k = |X|$. Then equivalently we prove that $Fn(k, 2)$ has the CCC. If $k \leq \omega$ then $|Fn(k, 2)| \leq \omega$ so it holds the same for any antichain. So wlog we can suppose $k > \omega$. By contradiction, let $A := \{s_\alpha : \alpha < \omega_1\}$ antichain. Let's group the elements based on the cardinality, that is consider: $\forall n < \omega \ A_n := \{s \in A : |s| = n\}$. Since ω_1 is regular, let $n < \omega$ be such that $|A_n| = \omega_1$. Now, any element $s \in A_n$ can be identified with the unique couple in $k^n \times 2^n$ given by $(\langle dom(s) \rangle, \langle ran(s) \rangle)$ where $\langle dom(s) \rangle$ is the enumeration of $dom(s)$ increasing according to the order given by k and $\langle ran(s) \rangle$ is the enumeration of $ran(s)$ as pointwise image of $\langle dom(s) \rangle$. To make an example, if $s = \{(0, 1), (\omega_2, 1), (\omega, 0)\}$ we we'll get $((0, \omega, \omega_2), (1, 0, 1))$. For every $t \in 2^n$, let $D_t := \{p \in k^n : (p, t) \text{ identifies some element of } A_n\}$. Now, since $|A_n| = \omega_1$ and $2^n < \omega$, we have that for some $t \in 2^n$, $|D_t| = \omega_1$. Since ω_1 is regular, using the Δ -system lemma we can find a root, r , for D_t . Take s_1, s_2 whose range, once identified is t . Then $dom(s_1) \cap dom(s_2) = r$. Since they are incompatible they must do something different in r . The possibilities however are at most $2^{|r|} < \omega$, which contradicts $|D_t| = \omega_1$.

□